

Chapter 1

Introduction to Project Management

Introduction

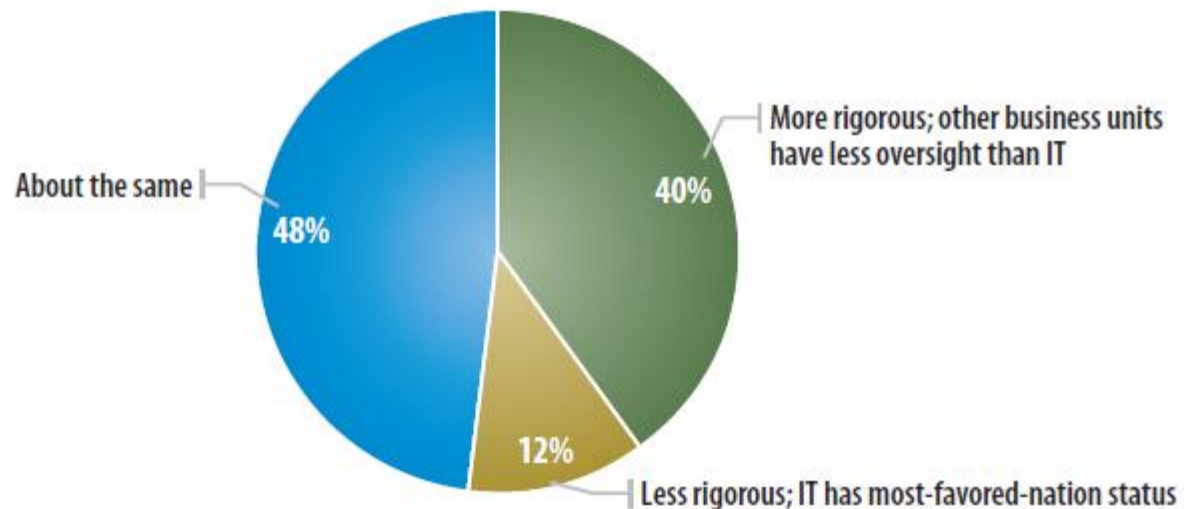
- The world as a whole spends nearly \$10 trillion of its \$40.7 trillion gross product on projects of all kinds
- More than 16 million people regard project management as their profession
- The overall information and communications technology market grew by 6 percent to almost \$3 trillion in 2010

Motivation for Studying Information Technology (IT) Project Management

- IT Projects have a terrible track record, as described in the “What Went Wrong?”

Scrutiny of IT Project Spending

Compared with other business units, the scrutiny placed on IT project spending at our organization is:



Advantages of Using Formal Project Management

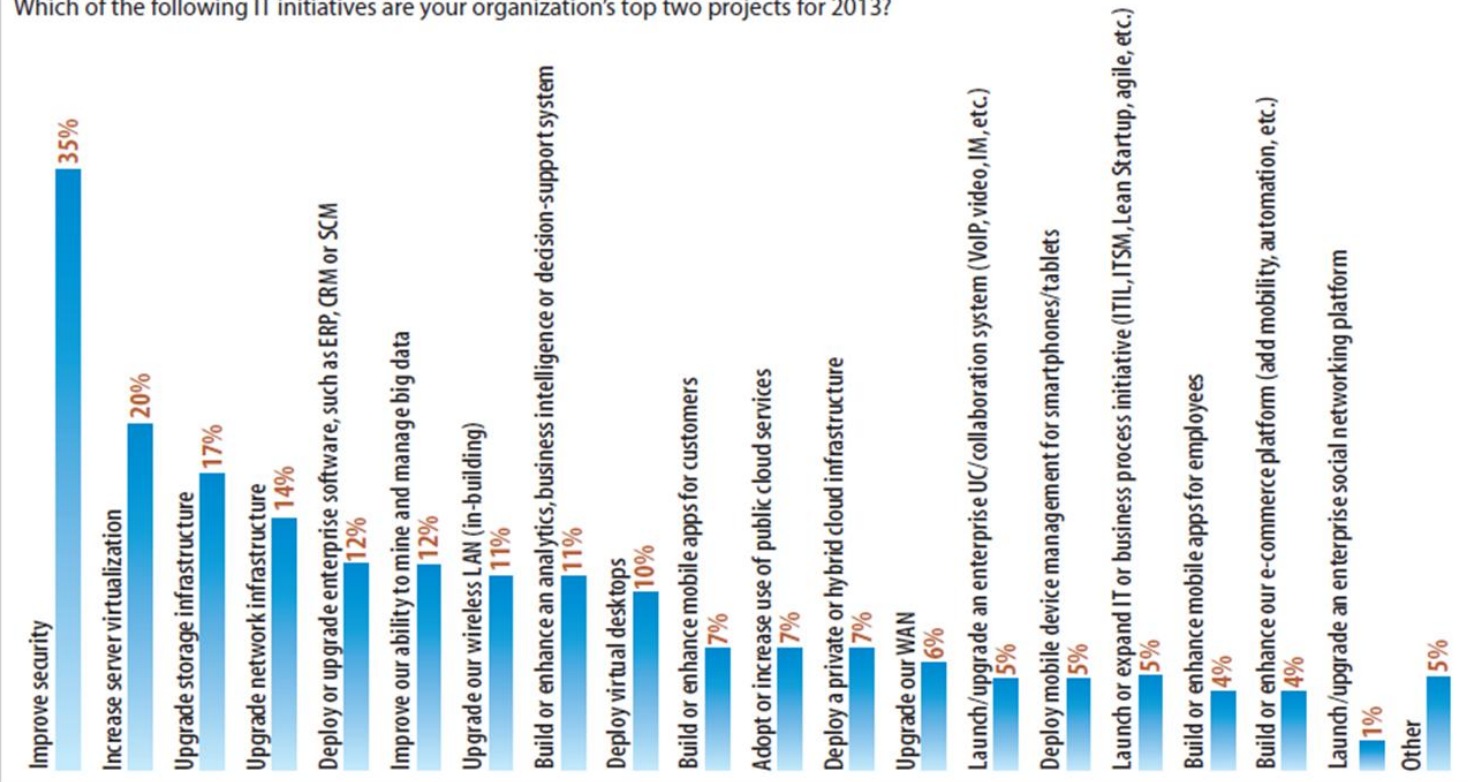
- Better control of financial, physical, and human resources
- Improved customer relations
- Shorter development times
- Lower costs
- Higher quality and increased reliability
- Higher profit margins
- Improved productivity
- Better internal coordination
- Higher worker morale

What Is a Project?

A **project** is “a temporary endeavor undertaken to create a unique product, service, or result” (PMBOK® Guide, Fifth Edition, 2012)

Top IT Projects for 2013

Which of the following IT initiatives are your organization's top two projects for 2013?



Note: Two responses allowed

Data: InformationWeek 2013 IT Spending Priorities Survey of 513 business technology professionals, March 2013

R6840513/2

Project Attributes

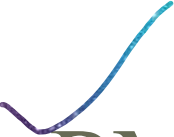
- A project
 - has a unique purpose
 - is temporary
 - is developed using progressive elaboration
 - requires resources, often from various areas
 - should have a primary customer or sponsor
 - The **project sponsor** usually provides the direction and funding for the project
 - involves uncertainty

Project and Program Managers

- **Project managers** work with project sponsors, project team, and other people involved in a project to meet project goals
- **Program:** group of related projects managed in a coordinated way to obtain benefits and control not available from managing them individually (PMBOK® Guide, Fifth Edition, 2012)

Program and Project Portfolio Management

- A **program** is “a group of related projects managed in a coordinated way to obtain benefits and control not available from managing them individually” (PMBOK® Guide, Fifth Edition, 2012)
- A **program manager** provides **leadership and direction** for the project managers heading the projects within the program

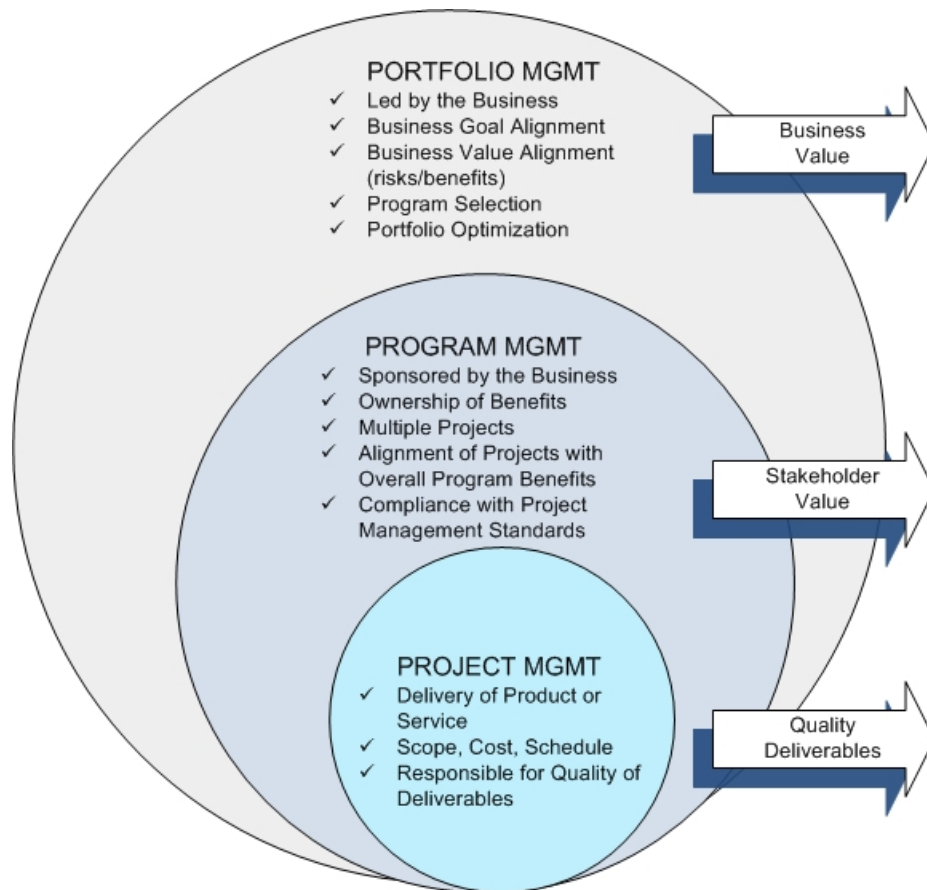


PM Network: What's in a Name?

■ Programs vs. Projects

- Should there be a difference?
- What are the problems with labeling a program as a large project?
- Are different skills needed to be a program manager compared to a project manager?

Project Portfolio Management



Organizations group and manage projects and programs as a portfolio of investments that contribute to the entire enterprise's success

Sample Project Portfolio Approach

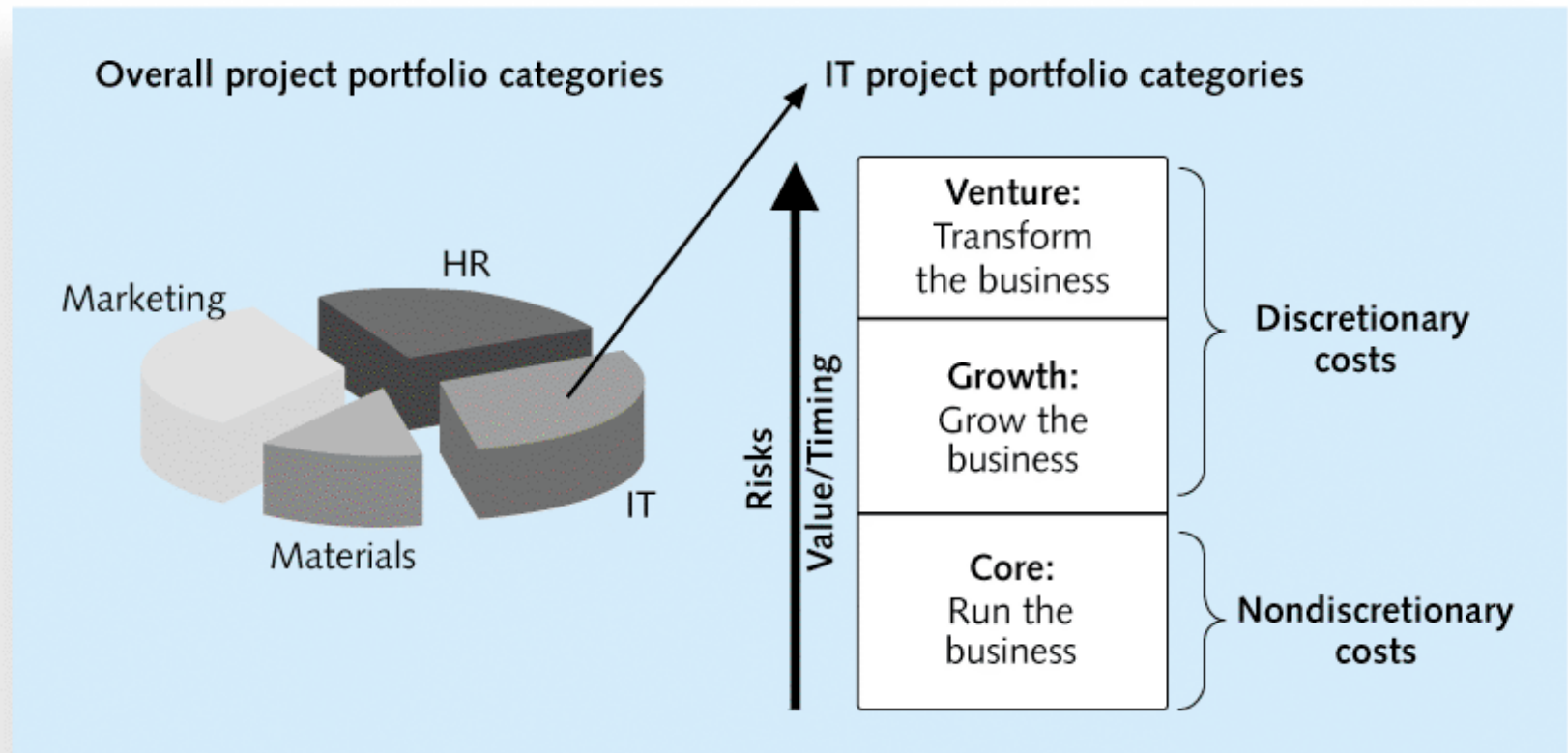
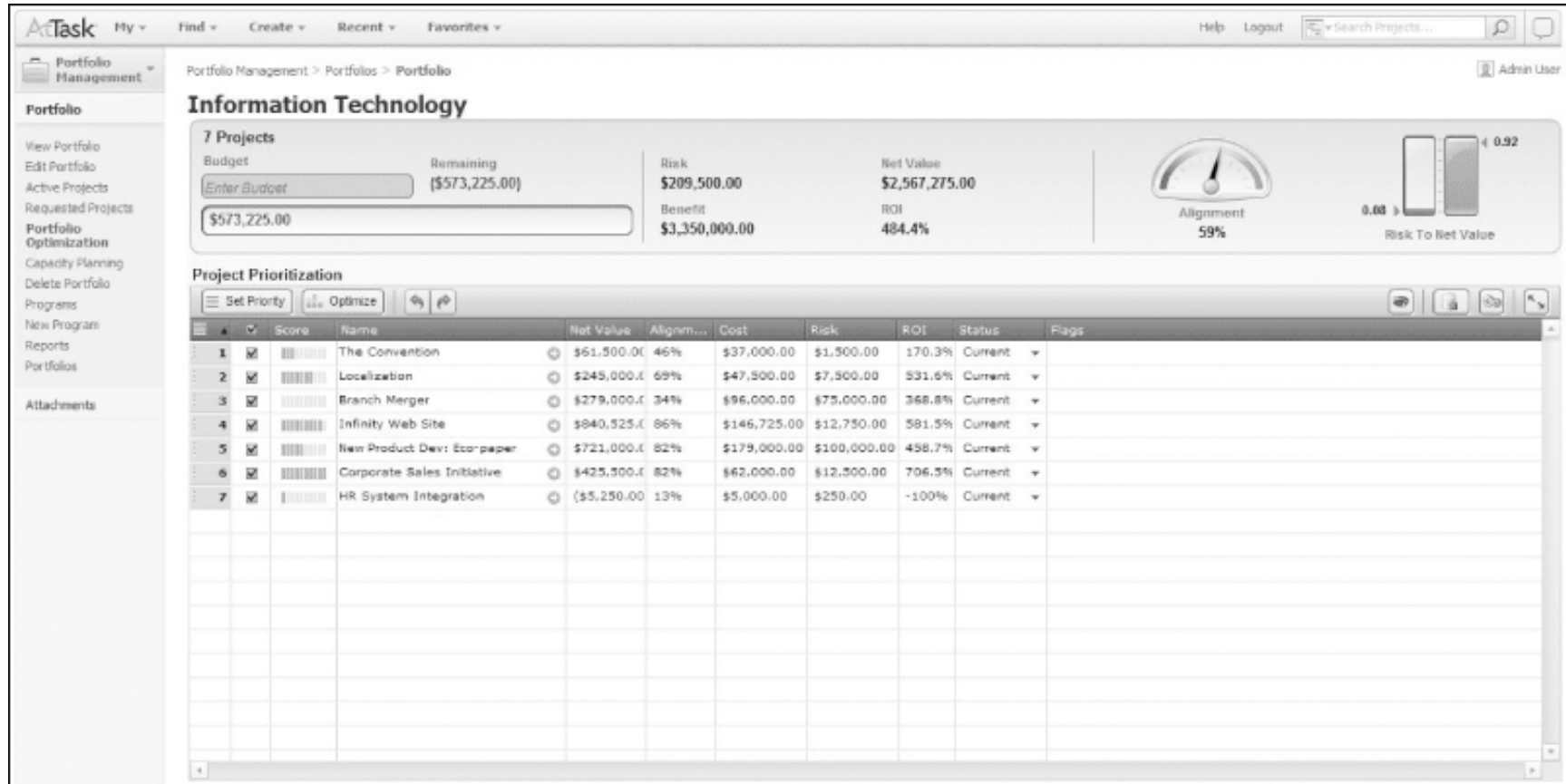
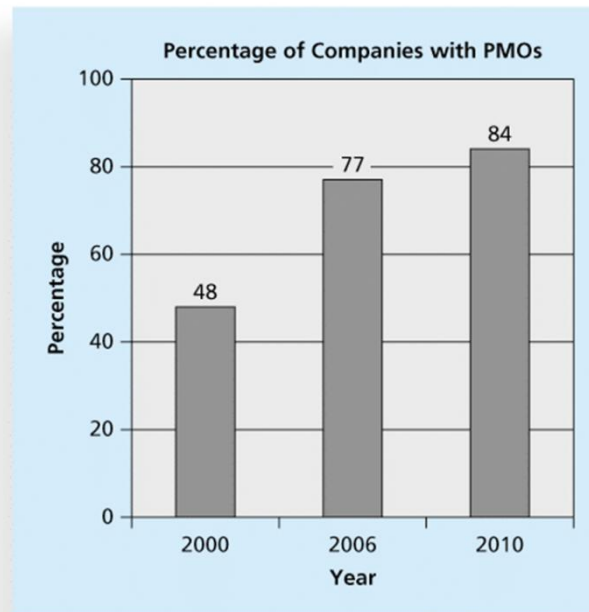


Figure 1-5. Sample Project Portfolio Management Screen Showing Portfolio Optimization



Project Management Offices

- A **Project Management Office (PMO)** is an organizational group responsible for coordinating the project management function throughout an organization



PM Network - PMO 2.0

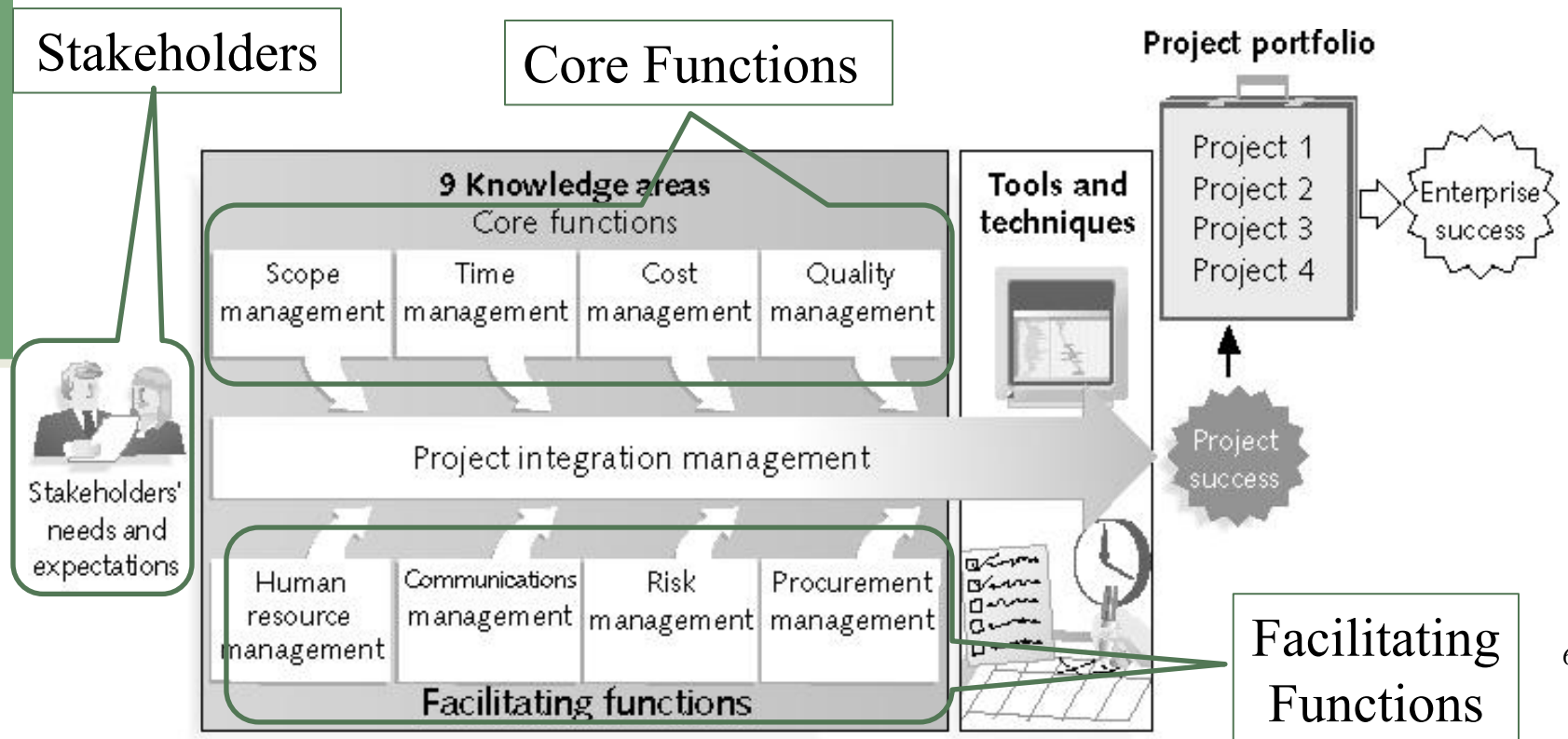
- Why do PMOs fail?
 - What is the primary reason cited for failure?
- How do you resurrect a failed PMO?
- How do you ensure longevity of a PMO?
- When should a PMO just be closed?

The Triple Constraint of Project Management



What is Project Management?

“the application of knowledge, skills, tools and techniques to project activities to meet project requirements” (PMBOK® Guide, Fourth Edition, 2012)

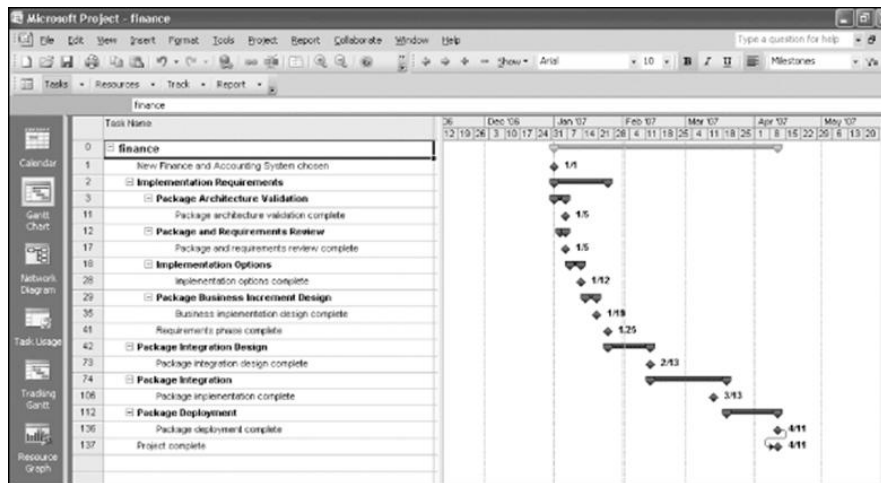


Project Management Knowledge Areas

- **Knowledge areas** describe the key competencies that project managers must develop
 - Core Functions
 - Facilitating Functions
 - Integration Function

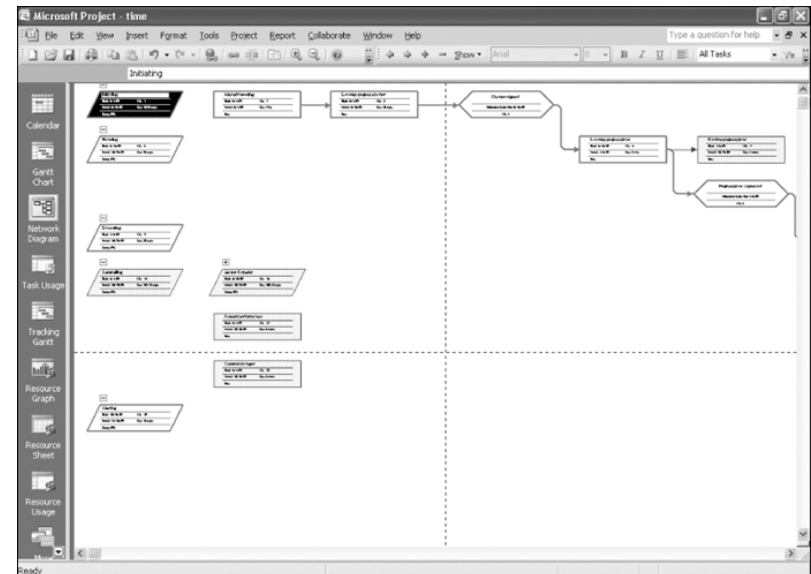
Project Management Tools and Techniques

- **Project management tools and techniques** assist project managers and their teams in various aspects of project management



Gantt chart

network diagram

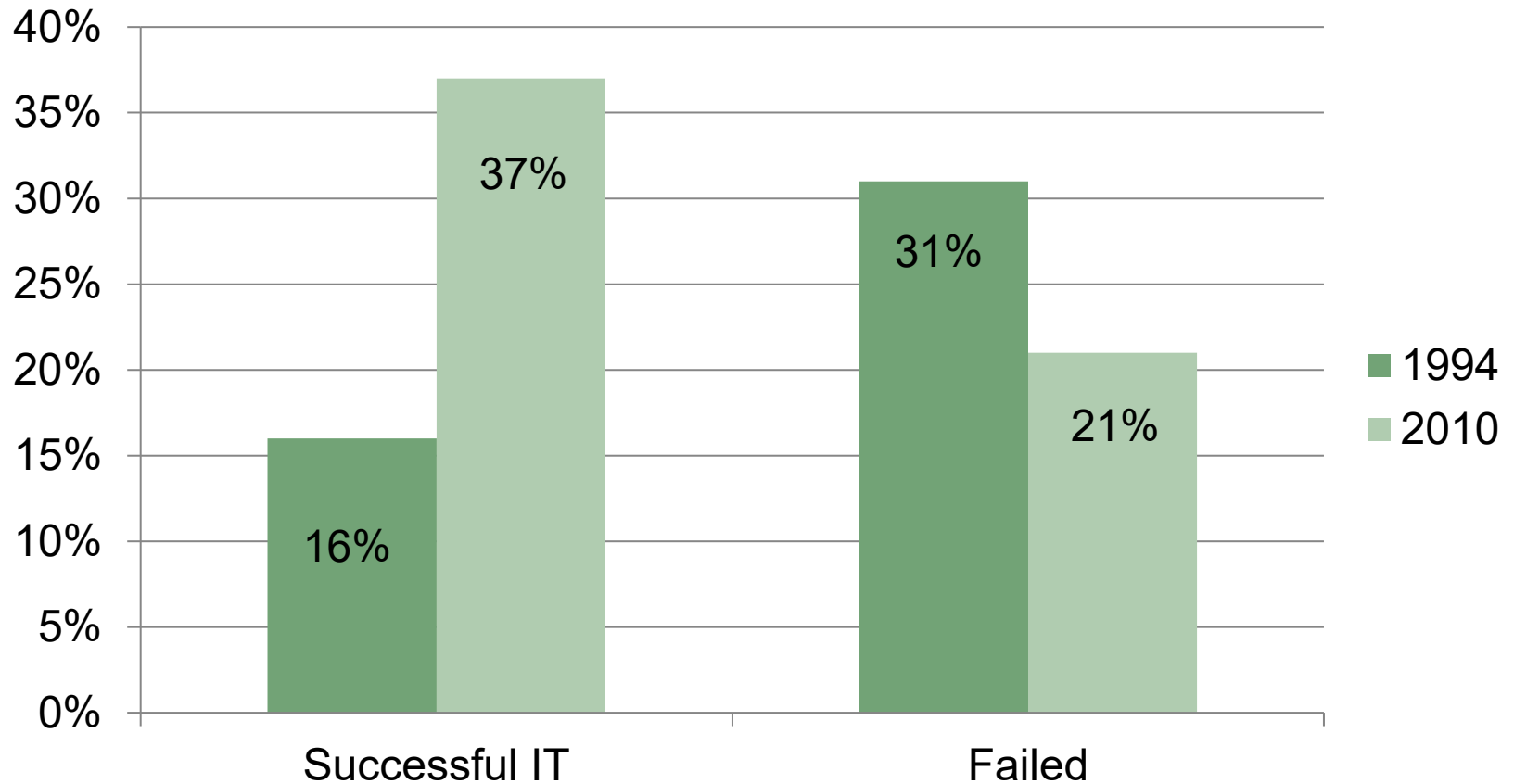


Project Stakeholders

- **Stakeholders** are the people involved in or affected by project activities

Who are the stakeholders in a project?

What Went Right? Improved Project Performance

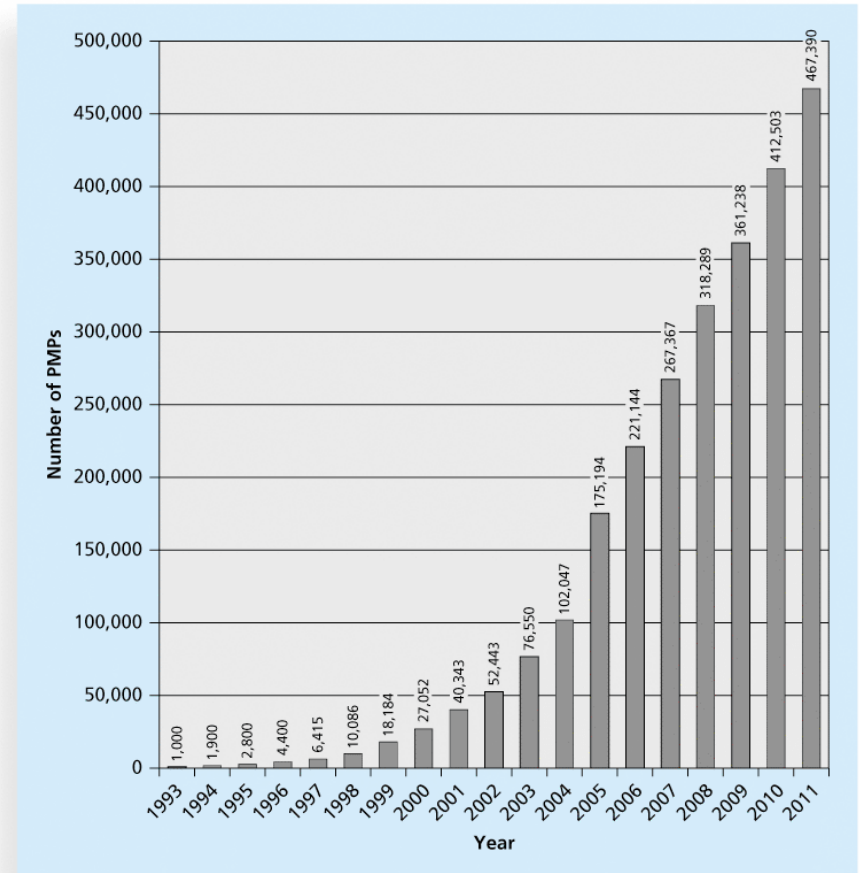


Why the Improvements?

Improved Project Performance

Growth in PMP Certification,
1993-2011

Coincidence?



Project Success

- There are several ways to define project success:
 - Triple Constraint
 - Customer/Sponsor Satisfaction
 - The results of the project met its main objective

What Helps Projects Succeed?*

1. User involvement
2. Executive support
3. Clear business objectives
4. Emotional maturity
5. Optimizing scope
6. Agile process
7. Project management expertise
8. Skilled resources
9. Execution
10. Tools and infrastructure

*The Standish Group, “CHAOS Activity News” (August 2011).

The Role of the Project Manager

- Job descriptions vary, but most include responsibilities like planning, scheduling, coordinating, and working with people to achieve project goals
- Remember that 97% of successful projects were led by experienced project managers, who can often help influence success factors

Suggested Skills for Project Managers

- The Project Management Body of Knowledge
- Application area knowledge, standards, and regulations
- Project environment knowledge
- General management knowledge and skills
- Soft skills or human relations skills

Ten Most Important Skills and Competencies for Project Managers

1. People skills
2. Leadership
3. Listening
4. Integrity, ethical behavior, consistent
5. Strong at building trust
6. Verbal communication
7. Strong at building teams
8. Conflict resolution, conflict management
9. Critical thinking, problem solving
10. Understands, balances priorities

Importance of Leadership Skills

- Effective project managers provide leadership by example
- A **leader** focuses on long-term goals and big-picture objectives while inspiring people to reach those goals
- A **manager** deals with the day-to-day details of meeting specific goals
- Project managers often take on the role of both leader and manager

Different Skills Needed in Different Situations

- Large projects
- High uncertainty projects
- Very novel projects

Careers for IT Project Managers

Job Categories	Total Current Employees Rank	Growth Rank
Big Data / Analyst	6	1
Business/Systems Analyst	3	3 (tie)
Database Admin / Analyst	7	3 (tie)
Networks / Security	5	5
Project Management	4	2
Software Development	2	6
Other IT Skills(Primarily Help Desk)	1	7

- In a 2012 survey, IT executives listed the “nine hottest skills” they planned to hire for in 2013
- Project management was second only to programming and application development

Nine Hottest Skills*

Skill	Percentage of Respondents
Programming and application development	60%
Project management	44%
Help desk/technical support	35%
Networking	35%
Business intelligence	23%
Data center	18%
Web 2.0	18%
Security	17%
Telecommunications	9%

*Source: Rick Saia, “9 Hot IT Skills for 2012,” Computerworld, September 26, 2011. ³⁰

The Project Management Profession

- The profession of project management is growing at a very rapid pace
- It is helpful to understand the history of the field, the role of professional societies like the Project Management Institute, and the growth in project management software

Global Issues

- Several global dynamics are forcing organizations to rethink their practices:
 - Talent development for project and program managers is a top concern
 - Good project portfolio management is crucial in tight economic conditions
 - Basic project management techniques are core competencies
 - Organizations want to use more agile approaches to project management
 - Benefits realization of projects is a key metric

Project Management Certification

- The Project Management Institute (PMI) is an international professional society for project managers with 380,000 members worldwide in 2012
 - **Project Management Professional (PMP)** has documented sufficient project experience, agreed to follow a code of ethics, and passed the PMP exam
 - **Certified Associate in PM (CAPM)** is achievable with less experience
- CompTIA offers another certification option
 - **CompTIA Project+** has less requirements but is not as well recognized as PMP

Ethics in Project Management

- **Ethics**, loosely defined, is a set of principles that guide our decision making based on personal values of what is “right” and “wrong”
- Project managers often face ethical dilemmas
- In order to earn PMP certification, applicants must agree to PMI’s Code of Ethics and Professional Conduct
- Several questions on the PMP exam are related to professional responsibility, including ethics

Project Management Software

- There are hundreds of different products to assist in performing project management
- Three main categories of tools:
 - Low-end tools
 - Midrange tools
 - High-end tools

Project Management Software

■ Various software includes:

- [BaseCamp](#)
- [Clarizen](#)
- [Collabtive](#) (open source)
- [dotProject](#) (open source)
- [OneDesk](#)
- [Genius Inside](#)
- [PlanBox](#)

Class Summary

- Project Definition
- Difference between project, program and portfolio
- Project Management Office
- Certification Opportunities
 - PMI/PMP
 - CompTIA